Lesson 3: Weave It

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

No preparation needed. Everything is collected with student helpers during the lesson. **No screens** are used. This week each student's **woven mat** is glued straight into their **STEAM journal**. Journals are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:
To weave strips of paper together to make a material that is stronger than the strips on their own.

Success Criteria:
I can:

- weave paper strips over and under.
- test how many pebbles my woven mat can carry.
- explain how my woven mat is different from loose strips.

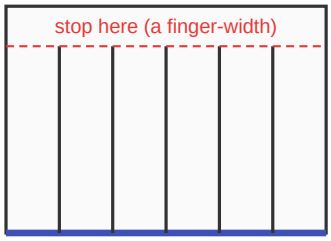
Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
A4 paper	2 sheets per student	1 sheet for the loom, 1 sheet for strips. Use coloured paper for the strips if there is any in the room.
Scissors	1 per student	Cutting slits in the loom
Black pebbles	About 15 per pair	The carry test
Sticky tape	A few pieces per table	Fixing loose strip ends (only if needed)
Glue sticks	1 per pair	Gluing the mat into journals
STEAM journals	1 per student	Shelves by the door

How to weave and test

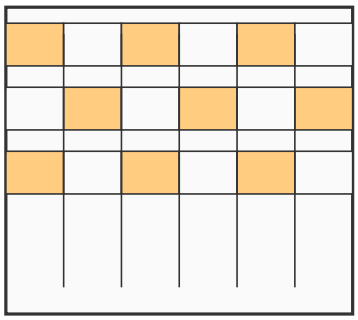
1. Fold and cut slits



Fold (cut from here)

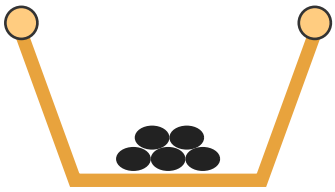
Cut 5 slits from the fold

2. Open and weave over, under



Next strip: under, over (swap)

3. Carry test



Lift the mat by two edges, just above the table.

Add pebbles 1 at a time.

- **Make the loom:** fold one A4 sheet in half (short edge to short edge). Starting from the fold, cut 5 slits, stopping **a finger-width from the edge**. Open it out. The uncut border holds the loom together.
- **Make strips:** fold the second sheet in half the long way, then in half again. Open it and tear along the 3 folds to make 4 long strips. Tear one more sheet between two students if they need extra.
- **Weave:** push the first strip **over, under, over, under** through the slits. Push the next strip **under, over, under, over** (the opposite). Slide each strip up close to the last one. 4 or 5 strips fill the loom.
- **Carry test:** first lay 4 **loose** strips side by side and put pebbles in the middle. Lift by two edges. The strips slide apart and the pebbles fall. Then do the same with the **woven mat**: lift it by two edges like a hammock, just above the table, adding 1 pebble at a time. Count how many it carries.

Key words

weave, over, under, strip, loom, pattern, strong, carry

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Ask: *"Last week, how did we make paper stronger?"* (By twisting strips into a rope.) Read the Learning Intention and Success Criteria aloud.
3 to 9 min	CIC hook	Lay 4 loose paper strips side by side on the floor and put 3 pebbles on them. Ask: *"If I lift these strips, will they carry the pebbles?"* Thumbs up or down. Lift them by two edges. The strips slide apart and the pebbles fall! Ask: *"How could we make these strips stay together without glue or tape?"* Take ideas. Then ask students to look closely at the sleeve of their jumper or shirt: tiny threads are joined together to make fabric. Explain: people have been **weaving** for thousands of years. Aboriginal peoples, including the Wurundjeri People, weave baskets and bags from grass fibres to carry food and other things. Today we will weave with paper.
9 to 14 min	Show how	Model the loom: fold, cut 5 slits from the fold, stop a finger-width from the edge, open. Model tearing strips. Model weaving the first strip (over, under) and the second strip (under, over), saying the pattern aloud with the class.
14 to 31 min	Make and weave	Helpers hand out 2 sheets and scissors to each student. Students make their own loom and strips, then weave. As students finish, pairs collect about 15 pebbles and do the carry test: loose strips first, then the woven mat. Roam with the questions below.
31 to 39 min	Record	Helpers hand out journals. Students glue their woven mat into their journal and write the date and the title **Weave It**. Underneath they copy and finish the sentences from the board.
39 to 45 min	Share and pack up	3 or 4 students share using the sentence stem. Finish with the key idea (below). Pack up: scraps into recycling, pebbles counted back into their container, scissors back, journals onto the shelves by the door.

Write this on the board

Loose strips carried ___ pebbles.

My woven mat carried ___ pebbles.

Sentence stem: **My woven mat is stronger than loose strips because _____.**

Questions to ask while roaming

- What happened to the loose strips when you lifted them?
- What is holding your woven strips together? There is no glue!
- Is your woven mat made of anything new, or just paper?
- Where have you seen weaving before? (baskets, clothes, hats, mats)

Support and extension

Who	What to do
Support	Cut the slits for students who find scissors tricky, or let them cut just 3 slits. Weave the first strip together, saying "over, under" aloud. Two woven strips is a success.
Extension	Fast finishers (often Year 2) make a colour pattern (for example 2 colours, AB AB), weave the strips tighter, or fold the strip ends neatly onto the back. Can a tighter weave carry more pebbles? Write one sentence explaining why.

Key idea to finish on: Loose strips fall apart. When we **weave** them over and under, they hold each other in place and become one strong mat that can carry things. Same paper, combined in a new way, just like twisting last week. Next week we will **mix** two materials to make something new.

Notes for the teacher

- The most common mistake is cutting slits right to the edge, which cuts the loom apart. Show "stop a finger-width from the edge" clearly. If a loom falls apart, tape the border back together on the back.
- If strips keep sliding out, have students push them close together, or tape the strip ends on the back.
- Woven mats do not have to be finished to be glued in. A half-woven mat still shows the learning.
- Lifting happens over the table, just above the surface, so pebbles do not fall far.
- Please leave a short note about how far the class got and any students who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2S01:** how familiar products are designed and produced by people to meet needs (elaboration: Wurundjeri woven baskets from grass fibres).
- **Design and Technologies VC2TDE2C04:** explore the characteristics and properties of materials used to create designed solutions.
- **Maths:** repeating patterns (over, under), counting a collection, comparing quantities.
- **Art:** colour and pattern in weaving.